

AI-Based Image-to-Recipe Generation System Using CNN and Sequence Modeling

Project Proposal — Phase 1 (Idea Submission)
Project-Based Placement Readiness Program

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Abstract

Recipe discovery today is largely dependent on textual search, requiring users to already know the name of a dish or its key ingredients before a recipe can be retrieved. This creates a gap for situations where a user has a photograph of a food item but lacks the vocabulary or exact name to search for it. This project proposes an AI-based system that predicts ingredients directly from a food image using a Convolutional Neural Network (CNN) and subsequently generates a coherent, structured recipe using a sequence generation model. The system is designed as an end-to-end pipeline covering image upload, preprocessing, feature extraction, ingredient prediction, and recipe generation, with a lightweight web interface for user interaction. This document presents the problem statement, literature background, proposed architecture, technology stack, and expected outcomes identified during the Phase 1 idea-submission stage.

1. Introduction

Automated food recognition and recipe generation have gained increasing attention with advances in computer vision and natural language generation. Most existing consumer-facing recipe platforms rely on keyword or text-based search, placing the burden of identification on the user. An image-driven approach removes this dependency by allowing a photograph to serve as the query itself. This project explores how CNN-based visual recognition can be combined with a sequence generation model to move directly from a food image to a usable recipe, without requiring the user to supply any textual description.

2. Literature Review

A review was conducted covering three related areas: image-to-recipe generation techniques, CNN-based food recognition, and sequence generation models used for text/recipe synthesis.

2.1 Image-to-Recipe Generation

Prior work in this space has explored joint embedding spaces that align food images with their corresponding recipes, enabling cross-modal retrieval and, in some cases, generative recipe construction directly from visual input.

2.2 CNN-Based Food Recognition

Convolutional architectures have been widely applied to food image classification tasks, typically trained on large annotated food datasets to distinguish dish categories and, at a finer granularity, individual ingredients present in an image.

2.3 Sequence Generation Models

Sequence generation approaches, ranging from recurrent architectures to attention-based models, have been used to convert structured or semi-structured inputs (such as a predicted ingredient list) into fluent, human-readable text such as step-by-step cooking instructions.

2.4 Research Gaps Identified

The review highlighted a consistent limitation across existing platforms: a strong dependency on textual queries as the entry point for recipe discovery. Systems that accept an image as the sole input and directly output a complete recipe, rather than only a dish label or a ranked retrieval list, remain comparatively limited. This gap motivates the proposed system.

3. Problem Statement

Existing recipe search systems require users to already know what a dish is called or what ingredients it contains, which is not always possible from a photograph alone. There is a need for a system that can take a food image as direct input and autonomously predict its likely ingredients and generate a corresponding recipe, reducing reliance on prior textual knowledge.

4. Objectives

- To design a CNN-based model capable of extracting visual features from food images for ingredient prediction.
- To develop a sequence generation module that converts predicted ingredients into a structured, readable recipe.
- To design an end-to-end pipeline connecting image input, preprocessing, prediction, and recipe output.
- To build a simple web-based interface for image upload and recipe display.
- To evaluate the feasibility of the proposed pipeline as a functional prototype within the program timeline.

5. Proposed System Architecture

The proposed system follows a linear processing pipeline, illustrated conceptually below:

Image Upload → Preprocessing → CNN Feature Extraction → Ingredient Prediction → Recipe Generation
(Sequence Model) → Output Delivery to User Interface

5.1 Module Description

Module	Function
Image Upload	Accepts a food photograph from the user through the web interface.
Preprocessing	Resizes, normalizes, and cleans the input image for model consumption.
CNN Feature Extraction	Extracts visual features and predicts probable ingredients present in the image.
Recipe Generation	Converts the predicted ingredient set into a coherent, step-by-step recipe using a sequence generator.
Output Delivery	Displays the generated recipe to the user through the web front end.

5.2 User Interaction Flow

The user interacts with the system through a minimal web interface: an image is uploaded, the backend pipeline processes it and returns the predicted ingredients along with a generated recipe, which is then rendered back to the user without requiring any manual text input at any stage.

6. Technology Stack

Layer	Technology
Programming Language	Python
Deep Learning Framework	PyTorch
Backend / API	Flask
Image Processing	OpenCV
Frontend	HTML, CSS, JavaScript

7. Expected Outcomes

The project is expected to produce a working prototype pipeline that accepts a food image and returns a generated recipe without any textual input from the user, along with supporting technical documentation and an evaluation of the pipeline's feasibility within the scope of the program.

8. Project Scope

The current scope covers the design and prototyping of the image-to-recipe pipeline: ingredient prediction from a single food image and generation of a corresponding recipe. Multi-dish images, portion/nutritional estimation, and multilingual recipe output are considered outside the scope of the present phase.

9. Future Enhancements

- Support for multi-ingredient, multi-dish images in a single photograph.
- Nutritional information estimation alongside the generated recipe.
- Personalized recipe suggestions based on dietary preference or restriction.

References

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